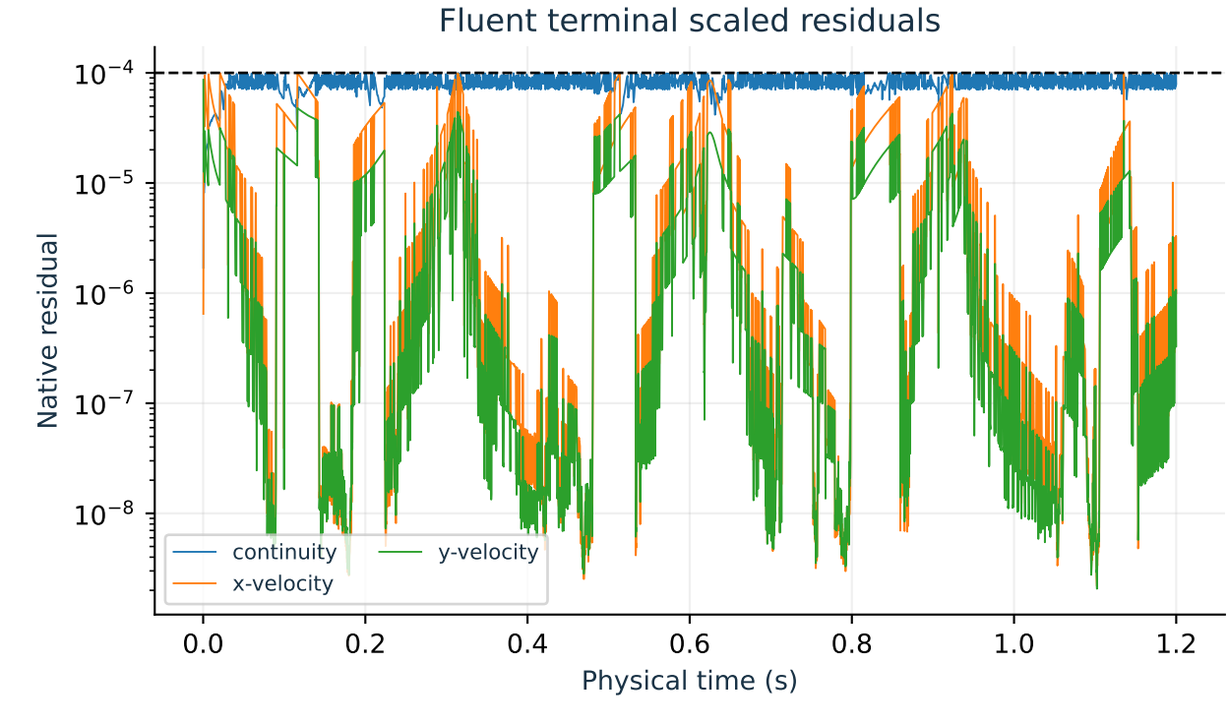
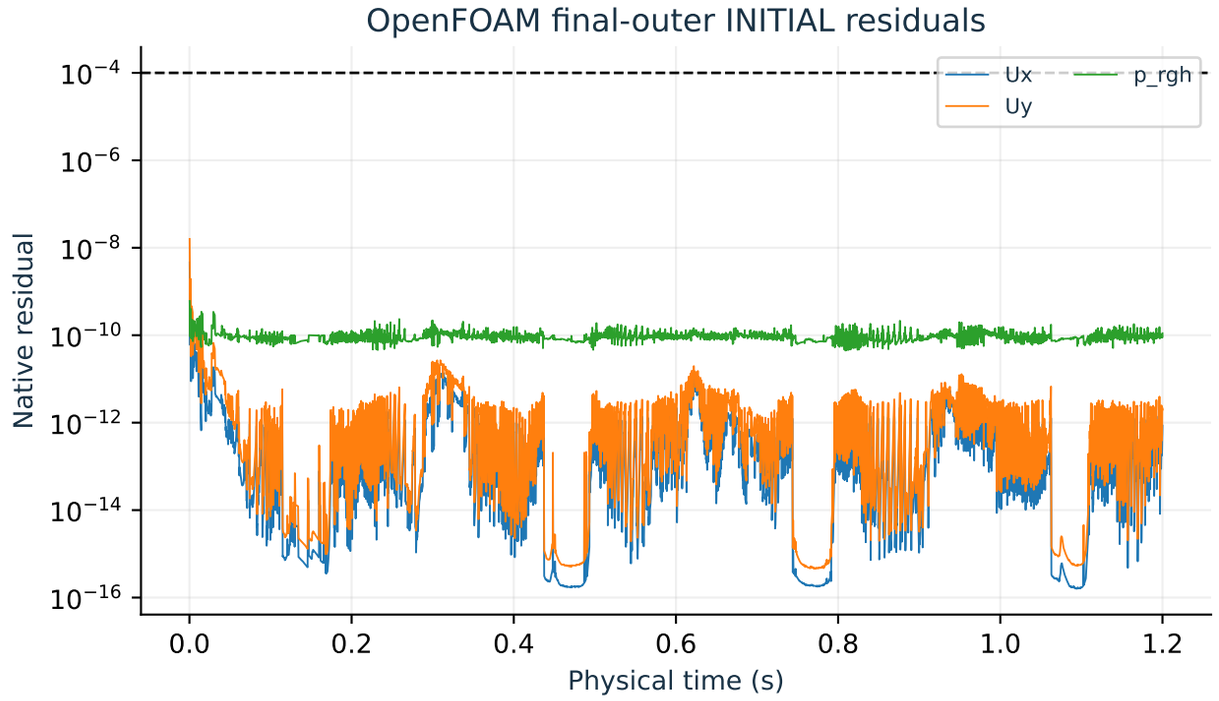
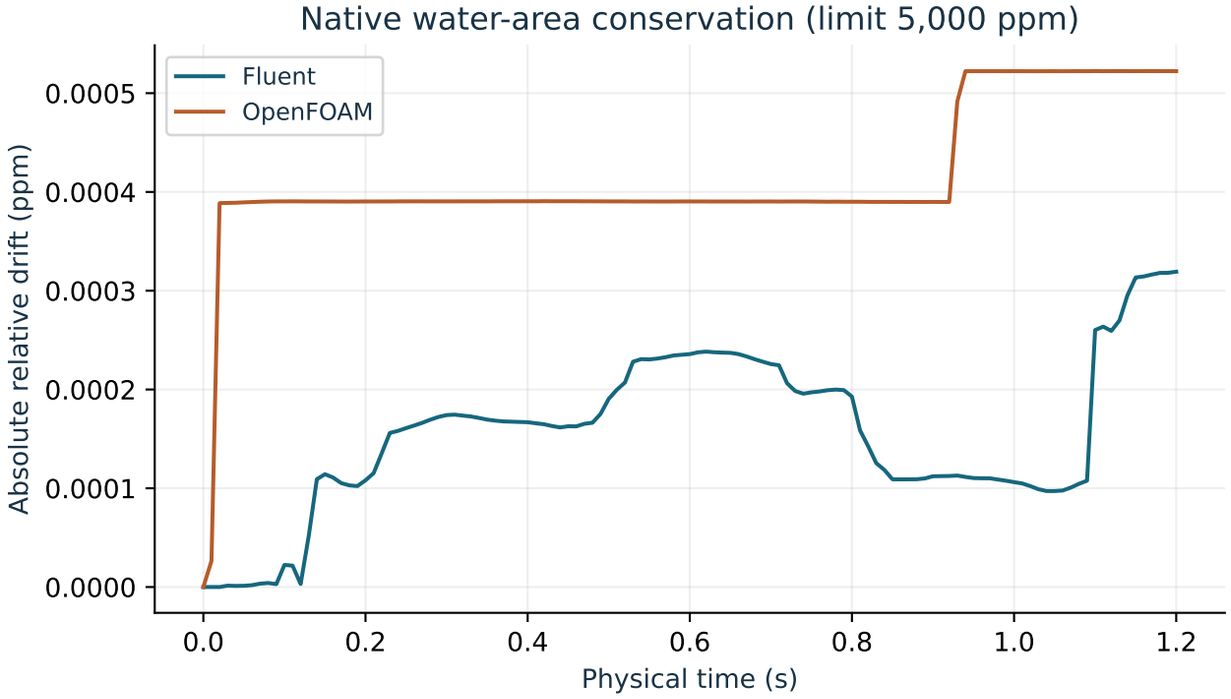
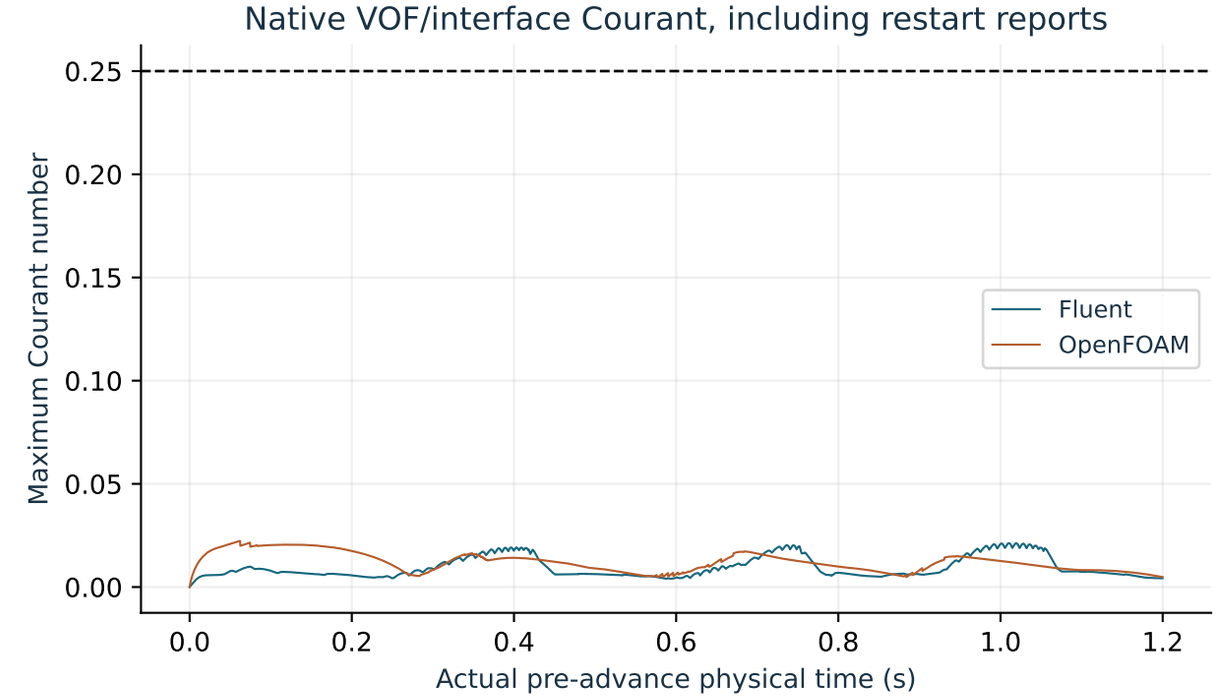
Accepted 4,800-step trajectories | native numerical evidence









Residual norms differ. Saved fraction bounds, clocks and wall impermeability pass; numerical convergence is not physical validation.